

Introduction to Large Language Models

1. What Are Large Language Models?

Large Language Models (LLMs) are deep learning systems trained on massive text corpora to understand and generate human language. They are based on the Transformer architecture, introduced by Vaswani et al. in 2017, which relies entirely on self-attention mechanisms rather than recurrence or convolution. Modern LLMs such as GPT-4, Gemini, Llama, and Claude contain billions of parameters and are capable of performing a wide range of tasks — from translation and summarization to code generation and reasoning — without task-specific training.

2. Transformer Architecture

The core building block of every LLM is the Transformer block. Each block consists of a multi-head self-attention layer and a position-wise feed-forward network, connected via residual connections and layer normalization. Self-attention allows every token in a sequence to attend to every other token, capturing long-range dependencies that RNNs struggled with. Positional encodings are added to token embeddings to preserve sequence order information.

3. Pre-training and Fine-tuning

LLMs are pre-trained using self-supervised objectives on large unlabeled datasets. The most common objective is next-token prediction (causal language modeling), where the model learns to predict the next word given all preceding words. After pre-training, models are aligned with human preferences using techniques such as Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO). Fine-tuning on instruction datasets further improves the model's ability to follow user instructions accurately.

4. Key Capabilities

Capability	Description	Example Models
Text Generation	Fluent, coherent long-form text	GPT-4, Gemini
Summarization	Condense documents to key points	Claude, Llama 3
Code Generation	Write and debug source code	CodeLlama, Codex
Question Answering	Answer factual/analytical Qs	GPT-4o, Gemini Pro
Translation	Multilingual text translation	NLLB, Gemini
Reasoning	Multi-step logical reasoning	o1, Claude 3.5

5. Scaling Laws

Empirical scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) describe how model performance improves predictably as model size, dataset size, and compute budget grow. The Chinchilla paper showed that many early large models were undertrained relative to their parameter count. Optimal training requires scaling tokens and parameters proportionally, roughly 20 tokens per parameter. These laws guide practitioners in allocating compute budgets efficiently across training runs.

6. Limitations and Safety Considerations

Despite their impressive capabilities, LLMs exhibit several limitations. They can hallucinate plausible-sounding but factually incorrect information. They may reproduce biases present in training data. They lack persistent memory across conversations. Adversarial prompts can sometimes elicit undesired outputs. Researchers are actively working on grounding techniques, retrieval-augmented generation, constitutional AI, and interpretability methods to address these challenges.